



① What is an API?

API = A way for two programs to talk to each others.

Your Frontend / mobile app / browser doesn't talk directly to the database.

It talk to the backend through an API.

Analogy

You = Customer (Frontend / App)

menu = API

Waiter = HTTP

kitchen = Backend logic + Database.

You:

- ⊙ Don't go into kitchen
- ⊙ Don't talk to chef directly
- ⊙ You use the menu to place order

The menu:

- ⊙ Show what you can ask for
- ⊙ Define format of request
- ⊙ Define what you'll get back

② Why do we need API?

- ⊙ Frontend and backend are separate.
- ⊙ Mobile app & web app can use same backend
- ⊙ other server can use your backend.



② you can change backend logic without breaking Frontend (if API stays same).

③ What is a Web API?

A Web API:

① Uses HTTP

① Uses URL

① Usually Send/Receive JSON

Example:

Get /users

Post /users

Get /users/5

DELETE /users/5

④ What is REST?

REST = Representational State Transfer

Rest is set of rules/style for designing APIs.

Rest says?

① Use Proper HTTP methods

① Use clean URLs

① Work with resources (user, product, orders).

① Be stateless (server doesn't remember client state).



⑤ REST + HTTP methods

Action	HTTP	Example
Get all users	GET	GET /users
Get one user	GET	GET /users/5
Create User	POST	POST /users
Update User	PUT	PUT /users/5
Delete user	DELETE	DELETE /users/5

⑥ Analogy

- ⊙ GET = Read menu
- ⊙ POST = Place order
- ⊙ PUT = Change order
- ⊙ DELETE = Cancel order

⑦ What does Backend API do?

Typical Flow:

1. Receive Request
2. validate data
3. Apply business Logic
4. talk to Database
5. Return response

Example -

POST /users

- validate name, email
- Save in DB
- Return created User



(8)

Implementation Example

From FastAPI import FastAPI

app = FastAPI()

@app.get("/users/{user-id}")

def get_user(user_id: int):

return {"user-id": user_id, "name":
"Demo User"}

if client calls:

GET /users/5

Response: JSON

```
{  
  "user-id": 5,  
  "name": "Demo User"  
}
```

(9) What does Stateless mean
in REST?

Stateless = Each request is independent

Server does not remember:

⊙ Who are you

⊙ What you did before

So server request must contain:

⊙ Auth info (token)

⊙ Needed data

Analogy

ATM machine:

⊙ It does not remember you

⊙ Every time you insert
card + PIN

⊙ Each operation is separated